Experiment No : 02

DATE

Group A

Title :-

Sorting employee salaries using selection sort and bubble sort algorithms.

Objectives :-

- To write a python program that sorts employee salaries (floating point numbers) using selection sort and bubble sort.
- To display the top five highest salaries after sorting.
- To analyze and compare the time and space complexity of both sorting algorithms.

Learning Objectives :-

By the end of this experiment students will be able to :

1) Understand the working principles of selection sort and bubble sort.
2) Implement sorting algorithms in python.
3) Compare the efficiency of sorting techniques.
4) Apply sorting in real-world applications like payroll systems

Learning Outcomes :-

After completing this experiment, students will be able to :

- Write program to sort data using basic sorting algorithm.
- Demonstrate understanding of sorting efficiency.
- Extract and display top salaries from sorted data.
- Analyze the time complexity ($O(n^2)$) and space complexity of the implemented algorithms.

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Theory :-

1] Selection Sort :-

- Divides the list into sorted and unsorted parts.
- Repeatedly selects the minimum element from the unsorted part and places it at the beginning.
- Time Complexity : $O(n^2)$
- Space Complexity : $O(1)$

2] Bubble Sort :-

- Repeatedly compares adjacent element and swaps them if they are in the wrong order.
- After each pass, the largest element "Bubble up" to its correct position
- Time Complexity : $O(n^2)$
- Space Complexity : $O(1)$

Comparison :-

- Both algorithm are simple but inefficient for large datasets.
- Both have the same worst-case complexity, but selection sort makes fewer swaps.

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